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# Relative Clause Attachment by L1 Vietnamese-L2 English Bilinguals: Effects of Prosody

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- Consider this sentence in English:

Jimmy met the brother of the engineer who has a smart dog.

Noun Phrase 1 Noun Phrase 2 Relative Clause

- Focus on attachment ambiguity: high attachment (HA) and low attachment (LA)
- Cross-linguistic mismatch in native speakers' default preference: English (LA) vs. Spanish/French (HA) (Fodor 2002; Cuetos & Mitchell 1988)
- Default attachment may reflect implicit prosody (Fodor 1998, 2002)
  - Readers assign a default prosodic contour during silent reading, which guides syntactic parsing and interpretation.
- Auditory stimuli may yield more consistent interpretations

- Prosodic phrasing affects attachment:
  - Break after NP1 → LA preferred: NP2 + RC in same Intonational Phrase (IP)
    - Jimmy met [the brother] // [of the engineer who has a smart dog].
  - Break after NP2 → HA preferred: RC in separate IP
    - Jimmy met [the brother of the engineer] // [who has a smart dog].
- L2 learners' interpretation patterns may vary:
  - Transfer from L1 (e.g., Fernández 2003)
  - No clear preference (e.g., Felser et al. 2003)
  - Native-like preferences (e.g., Dussias 2003)
- L2 learners can use prosodic cues to resolve ambiguity. (e.g., Dekydtspotter et al. 2008, Goad et al. 2021, Jun 2003)

- However, most studies focus on L1–L2 pairs with similar ambiguous structures.
- Vietnamese RCs are more frequent in formal registers, influenced by contact with European languages. (Ngo 2021)
- RCs are introduced by the relative marker *mà*, which is usually not mandatory in RC constructions (Miller 1976)

| | | | | | |
|---|---|---|---|---|---|
| ông kĩ sư | | (mà) | đang | nghe | điện thoại |
| CL | engineer | REL | PROG | listen | phone |

‘the engineer who is answering the phone’

- RCs following a complex NP (equivalent to ‘NP1 of NP2’ in English) are marginal although possible in Vietnamese.

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  - **LA preferred:**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| (người) | bố | của | ông$_{i}$ | bác sĩ // | ông$_{i}$ | mà | đang | khóc |
| CL | father | of | CL | doctor | RSM | REL | PROG | cry |

'[$_{NP}$ the father of [$_{NP}$ the doctor, the one who is crying]]'

  - **HA preferred:**

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| (người$_{i}$) | bố | của | ông$_{j}$ | bác sĩ // | người$_{i/*j}$ | mà | đang | khóc |
| CL | father | of | CL | doctor | RSM | REL | PROG | cry |

'[$_{NP}$ the father of [$_{NP}$ the doctor]], the one who is crying'

  - **Ambiguous:** (unnatural)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| ông$_{i}$ | bố | của | ông$_{j}$ | bác sĩ // | ông$_{i/j}$ | mà | đang | khóc |
| CL | father | of | CL | doctor | RSM | REL | PROG | cry |

'the father of the doctor, the one who is crying'

- RCs following a complex NP (equivalent to 'NP1 of NP2' in English) are marginal although possible, but classifier repetition sounds unnatural. *Can Vietnamese learners recognize RC ambiguity in English?*
- Prosody may assist L2 learners in identifying and resolving ambiguity.
- Vietnamese native speakers can use prosody to resolve ambiguity in certain sentence types.

| | | | | | | |
|---|---|---|---|---|---|---|
| Khi | uống | bia | **không** | được | pha | đường. |
| when | drink | beer | **NEG/plain** | able | put | sugar |

- Khi uống bia // không được pha đường.
  'When drinking beer, don't put sugar (in it).'
- Khi uống bia không // được pha đường.
  'When drinking beer plainly, you can put sugar.'
- Khi uống bia không được // pha đường.
  'When you can't drink beer, put sugar.'

(Nguyễn 2002:181-2)

- However, this strategy is not available for Vietnamese RC structures.

- Our study focuses on how Vietnamese-speaking learners disambiguate English RCs, especially how they process prosodic cues when such ambiguity is infrequent in their L1.
- Our hypothesis:

  Learners may not recognize ambiguity in English *NP1 of NP2 RC* structures.

  → As a result, they may not attend to prosodic cues for disambiguation.

# Methodology

- Participant profile
  - 30 Vietnamese learners of English, divided into Intermediate and Advanced based on cloze test scores.
  - 19 English native speakers

| | **Vietnamese learners of English (n = 30)** | | **English native speakers (n = 19)** |
|---|---|---|---|
| | Intermediate (n=15) | Advanced (n=15) | |
| Proficiency range (out of 30) | 9 – 22 | 23 - 28 | 16 – 30 |
| Average score | 16.3 | 25.50 | 24.56 |
| Standard deviation | 4.27 | 1.87 | 4.85 |

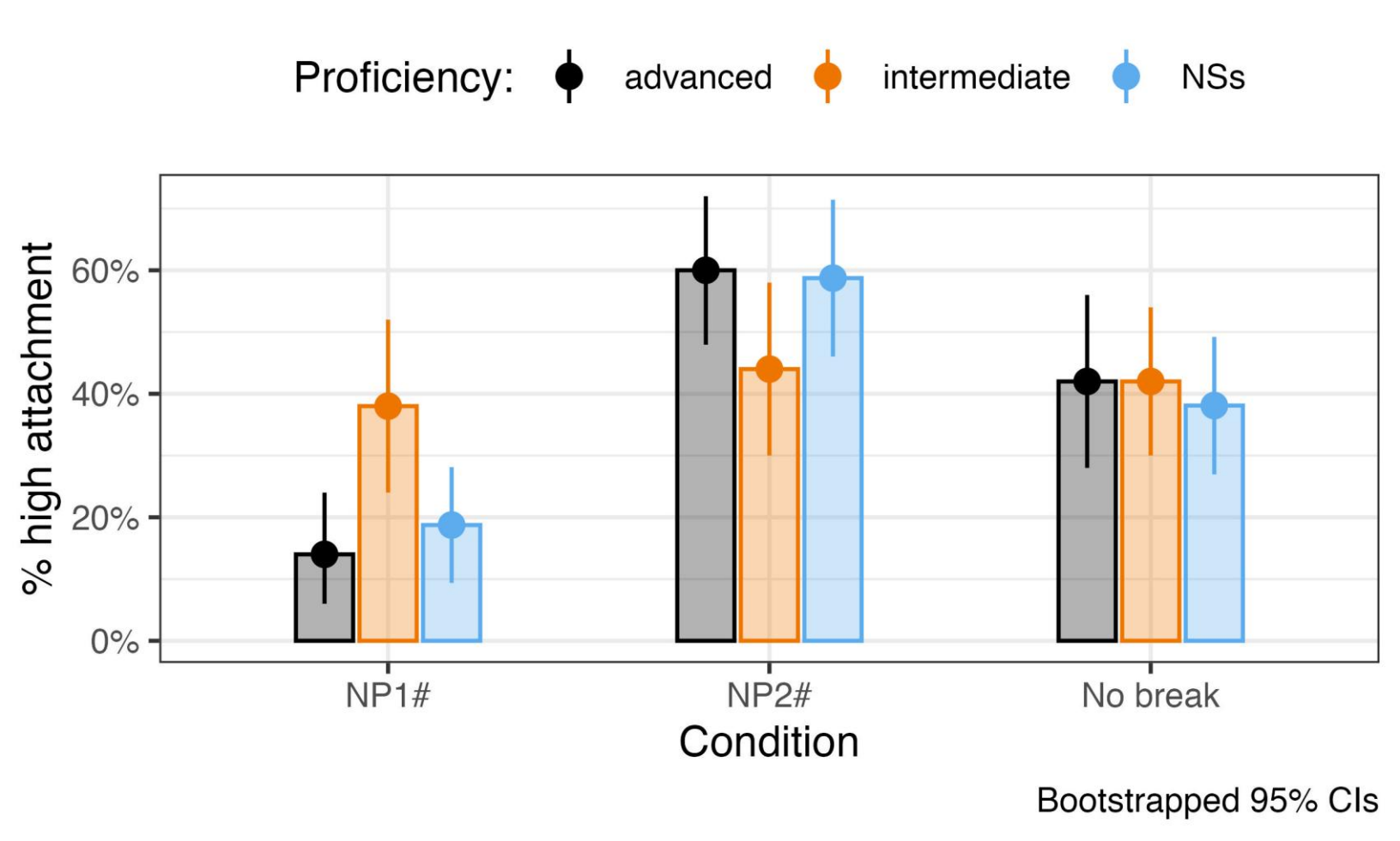
- Participant profile
- Stimuli: Auditory sentence interpretation task designed using Praat (Boersma & Weenink, 2025)
  - 30 sentences (10 target items + 20 unambiguous fillers)
  - Target structure: Subject + Verb + NP1 *of* NP2 + Subject RC
  - Three prosodic conditions:
    - No break: *Jimmy met the brother of the engineer who has a smart dog.*
    - Break after NP1: *Jimmy met the brother // of the engineer who has a smart dog.*
    - Break after NP2: *Jimmy met the brother of the engineer // who has a smart dog.*
  - Design: three versions of the task, each with one condition assigned to each target sentence

- Participant profile
- Stimuli
- Procedure (total duration: approx. 30 minutes)
  i. Background questionnaire
  ii. Sentence interpretation task:
    - Auditory presentation of testing items
    - Follow-up question: "*Who…?*" with two answer choices, NP1 (→ HA) or NP2 (→ LA)
  iii. Proficiency test:
    - Cloze test: 30 blanks in a reading passage; 4 options per blank (Xia et al. 2022)
    - Scoring: <23 = Intermediate; ≥23 = Advanced

- Participant profile
- Stimuli
- Procedure
- Predictions (by *Condition*)
  - Break after NP2 → more HA responses
  - Break after NP1 → fewer HA responses (i.e., LA preferred)
  - No break → slight LA preference (following Late Closure, Frazier 1979; and the default preference observed for English, Fodor 2002)
- Predictions (by *Group*)
  - English native speakers are sensitive to the use of prosodic cues for disambiguation.
  - L2 learners' sensitivity to the use of prosodic cues for disambiguation may be constrained by their proficiency (Advanced learners being more similar to Native speakers).

- Participant profile
- Stimuli
- Procedure
- Predictions
- Statistical analysis
  - Mixed-effects logistic regression using the lme4 package in R (R Core Team, 2025)
  - *Condition* and *Group* as main effects
  - Interaction between *Condition* and *Group*
  - Participant and item as random intercepts
  - Additional comparisons run with the lsmeans package (R Core Team, 2025)

# Results

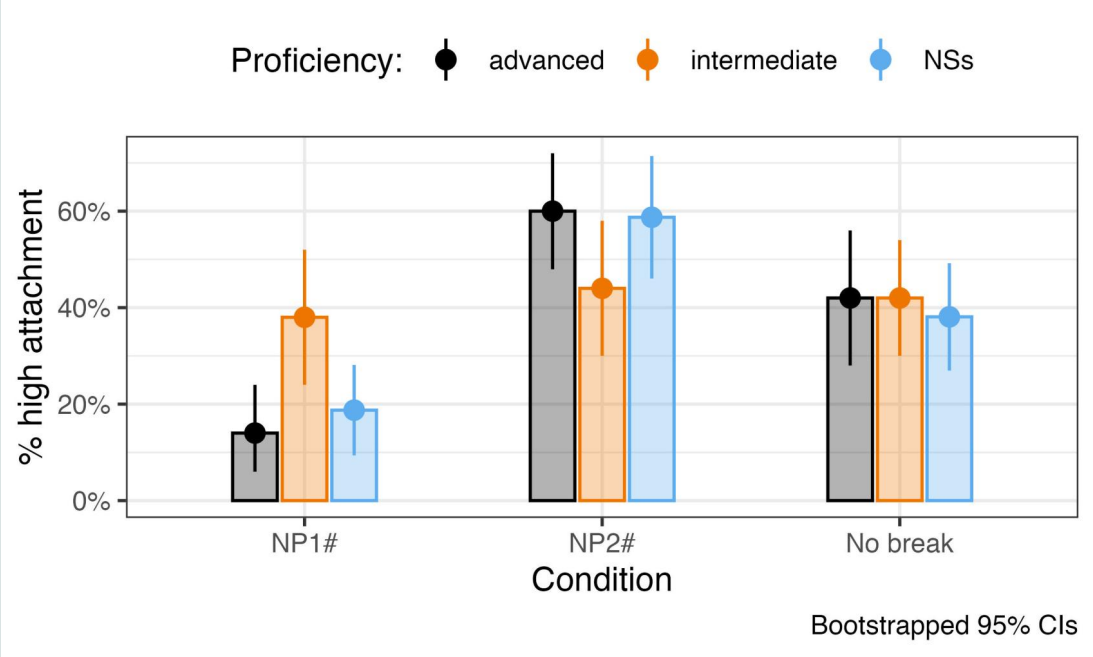

Proficiency:
advanced
intermediate
NSs
% high attachment
60%
40%
20%
0%
NP1#
NP2#
No break
Condition
Bootstrapped 95% CIs

# Results

- Main effect of *Condition*

(reference level = Break after NP1)

  - Break after NP1 is significantly different from Break after NP2 ($\hat{\beta}$ = 2.09, p < 0.0001)
  - Break after NP1 is significantly different from No Break ($\hat{\beta}$ = 1.05, p = 0.01)
  - Additional comparisons indicate that Break after NP2 is significantly different from No Break ($\hat{\beta}$ = 1.04, p = 0.009)

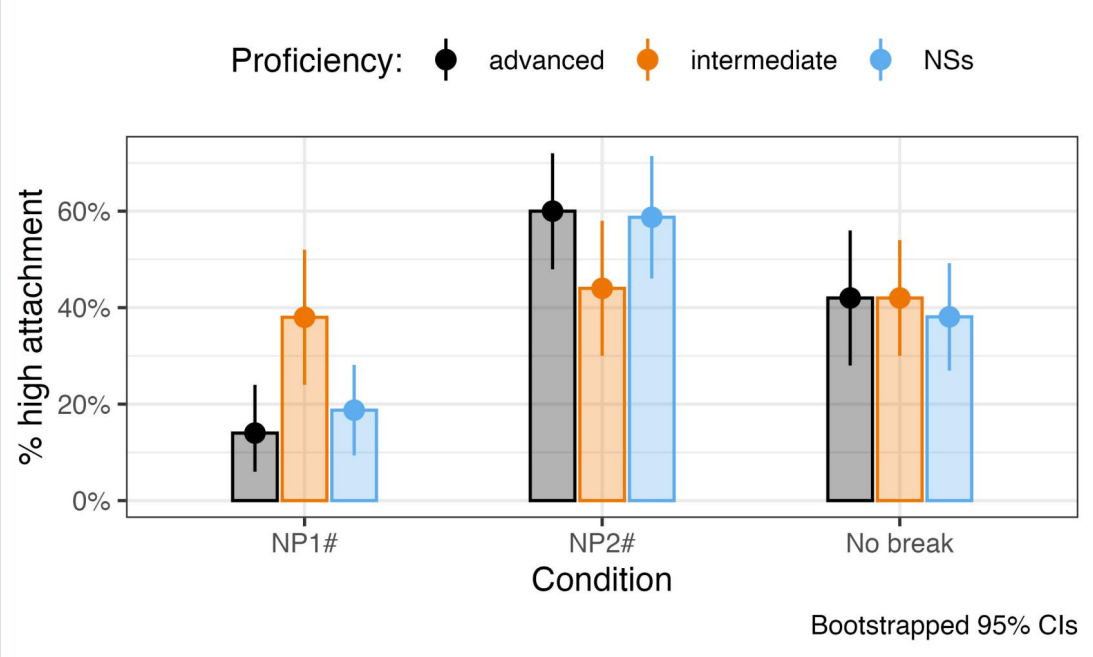


# Results

- Main effect of *Group*

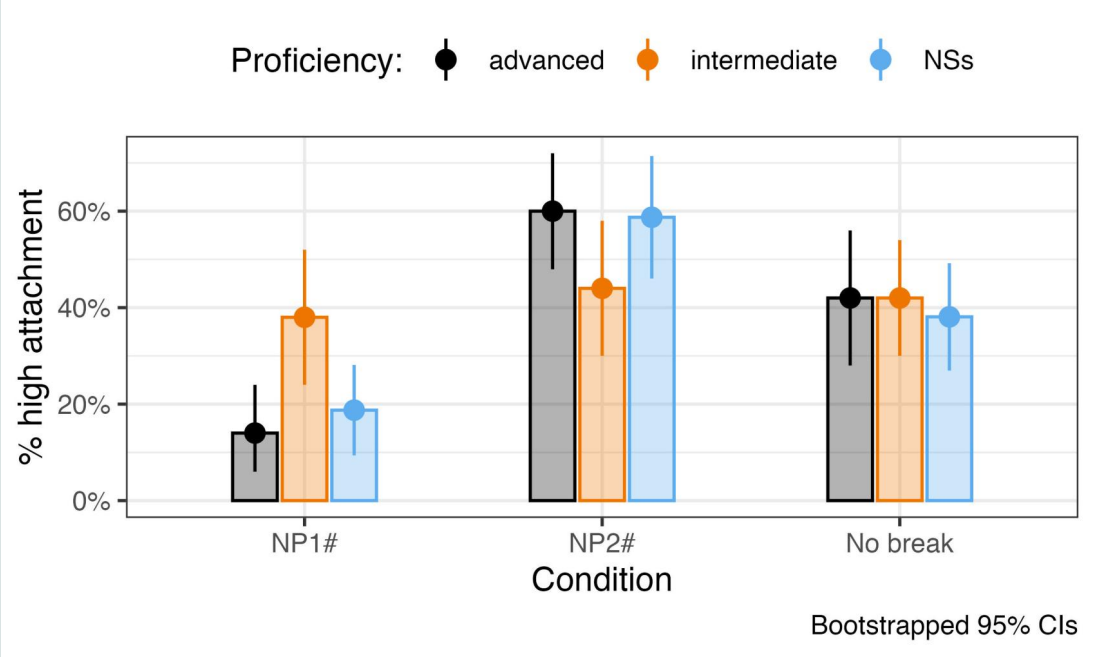
(reference level = Native speakers)

  - Native speakers are significantly different from Intermediate learners ($\hat{\beta}$ = 1.08, p = 0.03)
  - Native speakers are not significantly different from Advanced learners ($\hat{\beta}$ = –0.42, p = 0.47)
  - Additional comparisons indicate that Advanced learners are significantly different from Intermediate learners ($\hat{\beta}$ = –1.51, p = 0.009)



# Results

- Interaction between *Condition* and *Group*
  - Intermediate learners have fewer HA responses (than both Native speakers and Advanced learners) with Break after NP2 ($\hat{\beta}$ = –1.80, p = 0.003)



# Discussion and Conclusions

- Results for Native Speakers consistent with previous studies (e.g., Goad et al, 2021)
  - More HA responses when there is a break after NP2
  - Fewer HA responses when there is a break after NP1

- In the case of Native Speakers, results for No break consistent with the idea that LA is the default preference in English (following e.g., Fodor, 2002)

# Discussion and Conclusions

- Learners' results constrained by Proficiency
  - Advanced Learners performed similarly to Native Speakers
  - Intermediate Learners showed slight preference for LA in all conditions

- These results are consistent with previous research (e.g., Dekydtspotter et al. 2008; Fernández 2003; Goad et al. 2021) showing that more proficient learners are more sensitive to cues for disambiguation

# Discussion and Conclusions

- Intermediate learners' insensitivity to prosodic cues is puzzling
- Two possibilities (not mutually exclusive):
  - Intermediate learners are able to identify the ambiguity in the target sentences (but are unable to resolve it with prosodic cues)
  - Intermediate learners favour a local interpretation for ambiguous structures regardless of prosodic cues (following the *Late Closure* principle; Frazier, 1979)

# Discussion and Conclusions

- In summary, the data suggest:
  - Both Native Speakers and Learners can use prosodic cues to resolve ambiguities
  - Learners' performance is conditioned by their proficiency level, with advanced learners not being statistically different from native speakers
  - Preferences (in the case of intermediate learners) may also be influenced by principles such as Late Closure

# Discussion and Conclusions

- Future work
  - Learners' awareness of RC ambiguity in English
  - Production data: do learners produce disambiguating prosody?
  - Training effects: can prosody-based instruction improve interpretation?

# THANK YOU! XIN CẢM ƠN!

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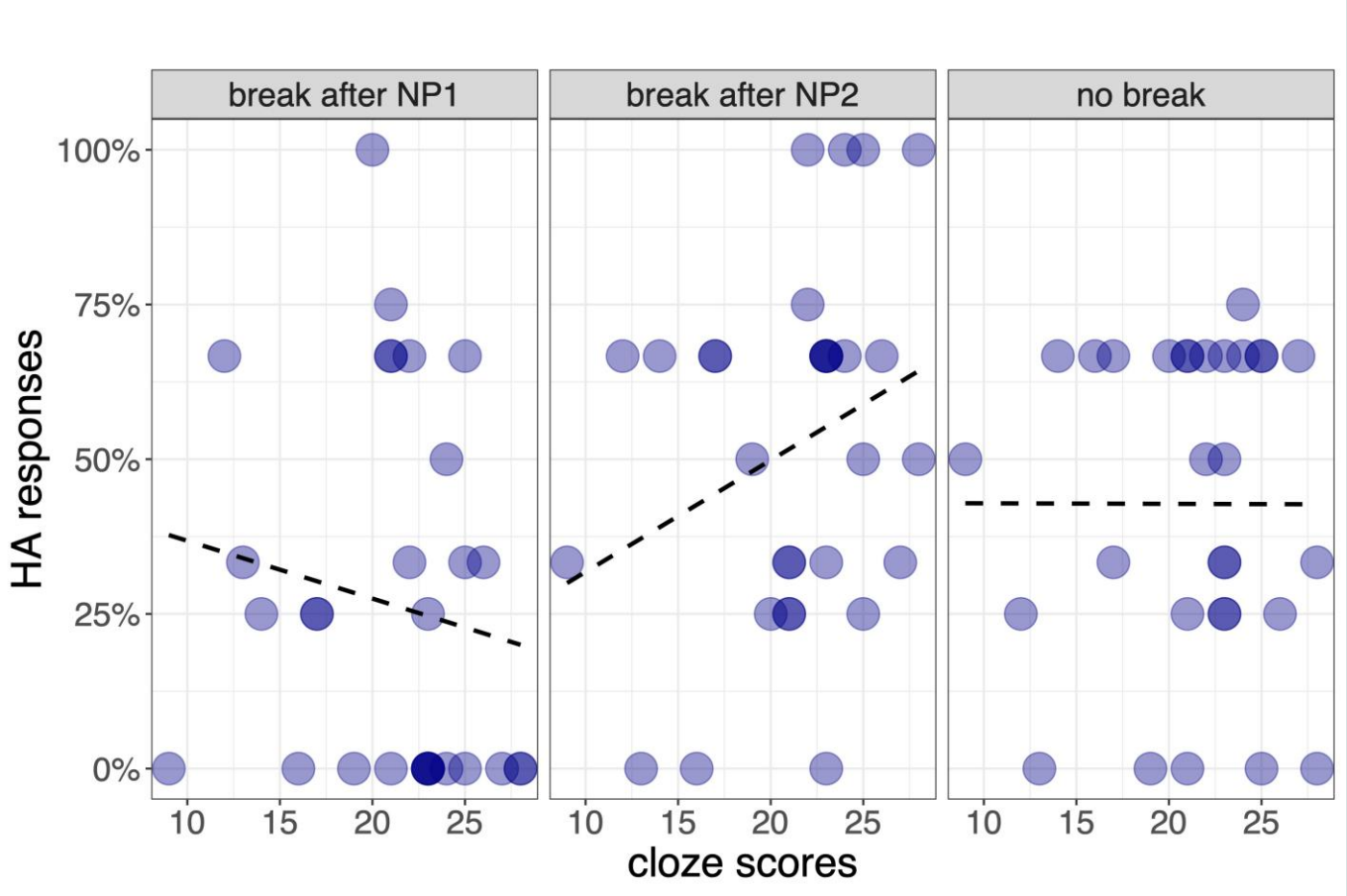
# Appendix A

## Target stimuli

1. Jimmy met the brother of the engineer who has a smart dog.
2. Anna yelled at the daughter of the actress who was waiting outside.
3. Adam gave money to the assistant of the dentist who plays golf.
4. Katie is dating the cousin of the doctor who enjoys football.
5. Bob found the uncle of the manager who went on a vacation.
6. Henry had dinner with the sister of the lawyer who has a red car.
7. Kim argued with the son of the journalist who speaks five languages.
8. George recognized the secretary of the director who was divorced.
9. Nora picked up the mother of the architect who likes banana bread.
10. Rachel was worried about the aunt of the child who has blue eyes.

# Appendix B

Preferences plotted by cloze test results



# Appendix C

## Statistical model

| | Estimate | Std. error | z value | p value |
|---|---|---|---|---|
| Intercept | –1.66 | 0.41 | –3.98 | < 0.001 |
| Condition (break after NP2) | 2.09 | 0.44 | 4.7 | < 0.001 |
| Condition (no break) | 1.05 | 0.43 | 2.41 | 0.01 |
| Proficiency (advanced) | –0.42 | 0.59 | –0.71 | 0.47 |
| Proficiency (intermediate) | 1.08 | 0.51 | 2.11 | 0.03 |
| Break after NP2:advanced | 0.46 | 0.69 | 0.66 | 0.5 |
| No break:advanced | 0.65 | 0.68 | 0.95 | 0.34 |
| Break after NP2:intermediate | –1.8 | 0.62 | –2.89 | 0.003 |
| No break:intermediate | –0.84 | 0.62 | –1.36 | 0.17 |